

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Determine all critical points of $f(x) = \sqrt{1-x^2}$.

$$\text{Note that } f'(x) = \frac{1}{2}(1-x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{1-x^2}}.$$

$$f'(x) = 0 \text{ when } -x = 0 \rightarrow x = 0.$$

$$f'(x) \text{ is undefined when } \sqrt{1-x^2} = 0 \rightarrow x^2 = 1 \rightarrow x = \pm 1.$$

Therefore, the critical values of $f(x)$ are $-1, 0$, & 1 .

2. Consider the function $f(x) = \frac{1}{\sqrt{x-2}}$

- a) Confirm that $x = 2$ either makes $f'(x)$ equal 0 or become undefined.

$$\text{We see that } f(x) = (x-2)^{-1/2} \text{ and } f'(x) = -\frac{1}{2}(x-2)^{-3/2} = \frac{-1}{(x-2)^{3/2}}.$$

Clearly $x = 2$ makes $f'(x)$ undefined.

- b) Explain why $x = 2$ is *NOT* a critical point of $f(x)$.

Remember that a critical point is defined as a point within the domain of $f(x)$ for which $f'(x) = 0$ or $f'(x)$ is undefined.

$x = 2$ is *NOT* a critical point of $f(x)$ because it is not in the domain of $f(x)$.

3. Find all critical points of $f(x) = \frac{x^2}{x^3+8}$.

$$\text{We calculate that } f'(x) = \frac{(x^3+8)2x - (3x^2)x^2}{(x^3+8)^2} = \frac{-x^4+16x}{(x^3+8)^2} = \frac{x(-x^3+16)}{(x^3+8)^2}$$

$$\text{Note that } f'(x) = 0 \text{ when } x(-x^3+16) = 0 \rightarrow x = 0 \text{ or } x = \sqrt[3]{16}$$

$$f'(x) \text{ is undefined when } x = -2.$$

Since $x = -2$ is not in the domain of $f(x)$ then only $x = 0$ & $x = \sqrt[3]{16}$ are critical points.

EXTENSIONS

4. Find all critical points of $y = x^2(x-1)^{2/3}$.

$$y' = 2x(x-1)^{2/3} + \frac{2}{3}(x-1)^{-1/3} x^2 = 2x(x-1)^{-1/3} \left[(x-1) + \frac{1}{3}x \right] = \frac{2x \left[\frac{4}{3}x - 1 \right]}{(x-1)^{1/3}}$$

$$y' = 0 \text{ when } 2x \left[\frac{4}{3}x - 1 \right] = 0 \rightarrow x = 0 \text{ \& } x = \frac{3}{4}$$

y' is undefined when $x = 1$.

Therefore, the critical values of y are 0 , $\frac{3}{4}$, & 1 .

5. Find all critical points of $f(x) = \sin|x|$.

$$\text{First, we note that } f(x) = \sin|x| = \begin{cases} \sin(x) & \text{if } x \geq 0 \\ \sin(-x) & \text{if } x < 0 \end{cases}$$

So, for $x \geq 0$ we know $f'(x) = \cos(x)$ and $\cos(x) = 0$ when $x = \frac{\pi}{2} + \pi k$ for $k = 0, 1, 2, \dots$

For $x < 0$ we know $f'(x) = -\cos(-x)$ and $-\cos(-x) = 0$ when $x = -\frac{\pi}{2} - \pi k$ for $k = 0, 1, 2, \dots$

We now need to check if $f(x)$ is differentiable at $x = 0$.

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{\sin|h| - \sin(0)}{h} = \lim_{h \rightarrow 0} \frac{\sin|h|}{h}$$

We will check this limit from both sides.

$$\lim_{h \rightarrow 0^+} \frac{\sin|h|}{h} = \lim_{h \rightarrow 0^+} \frac{\sin(h)}{h} = 1 \text{ and } \lim_{h \rightarrow 0^-} \frac{\sin|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-\sin(h)}{h} = -1$$

This implies that $\lim_{h \rightarrow 0} \frac{\sin|h|}{h}$ DNE and therefore $f(x)$ is not differentiable at $x = 0$.

Therefore, we have critical points of

- $x = \frac{\pi}{2} + \pi k$ for $k = 0, 1, 2, \dots$
- $x = -\frac{\pi}{2} - \pi k$ for $k = 0, 1, 2, \dots$
- $x = 0$

SUMMARY POINT: We need to understand what we need to look for when finding critical values. We are looking for

- (1) A value of x that makes $f'(x) = 0$ or $f'(x)$ is undefined.
- (2) A value of x that is in the domain of $f(x)$.

6. Consider the function $f(x) = \frac{ax+b}{x^2-1}$ where a and b are real numbers. Find values of a and b such that $f(x)$ has a local extreme value of 1 at $x=3$.

In order to see where $f(x)$ has local mins and maxes we find $f'(x) = \frac{(x^2-1)(a) - (2x)(ax+b)}{(x^2-1)^2}$.

If there is to be a local max at $x=3$ then $x=3$ must correspond to a critical point of f and therefore make $f'(x)$ either equal to 0 or undefined. Clearly $x=3$ does not make $f'(x)$ undefined so we need to find values of a and b such that $f'(3)=0$.

So, we must have:

$$\begin{aligned} ((3)^2-1)(a) - (2(3))(a(3)+b) &= 0 \rightarrow 8a - 6(3a+b) = 0 \rightarrow 8a - 18a - 6b = 0 \\ -10a - 6b &= 0 \rightarrow -2(5a+3b) = 0 \rightarrow 5a+3b = 0 \end{aligned}$$

Since we want a local extreme value of 1 this means that we also need $f(3)=1$.

So, we also must have:

$$\begin{aligned} 1 &= \frac{a(3)+b}{(3)^2-1} \\ 8 &= 3a+b \end{aligned}$$

In order for both of these things to be true we must have values of a and b that satisfy the system

$$\begin{cases} 5a+3b=0 \\ 3a+b=8 \end{cases}$$

We solve this system and find that

$$\begin{aligned} \begin{cases} 5a+3b=0 \\ 3a+b=8 \end{cases} &\rightarrow \begin{cases} 5a+3b=0 \\ 9a+3b=24 \end{cases} \rightarrow \frac{9a+3b=24}{4a=24} \rightarrow -\frac{5a+3b=0}{4a=24} \rightarrow a=6 \rightarrow b=-10 \end{aligned}$$

This means that $f(x) = \frac{6x-10}{x^2-1}$ and

$$f'(x) = \frac{(x^2-1)(6) - (2x)(6x-10)}{(x^2-1)^2} = \frac{-6x^2+20x-6}{(x^2-1)^2} = \frac{-2(3x^2-10x+3)}{(x^2-1)^2} = \frac{-2(3x-1)(x-3)}{(x^2-1)^2}$$

We can clearly see that $f'(3)=0$ and so by Fermat's Theorem $f(x)$ must have a local min or max at $x=3$. We also clearly see that $f(3)=1$.

7. TRUE or FALSE: A function $f(x)$ must have an absolute maximum or minimum on $[a, b]$.

FALSE. If the function isn't continuous on this interval then it doesn't have to hit either an absolute maximum nor an absolute minimum. Consider a function on $[1, 10]$ that has a vertical asymptote at $x = 5$.

EXTENSION:

8. TRUE or FALSE: The value $y = -10$ cannot be the absolute maximum of a function $f(x)$.

FALSE. It is entirely possible to have an absolute maximum of -10. For example, consider the function $f(x) = -x^2 - 10$.

SUMMARY POINT: Remember that the Extreme Value Theorem (EVT) tells us that if we have a continuous function on $[a, b]$ then this function must obtain an absolute maximum and absolute minimum on this interval. These extremes could either happen at a critical point within the interval or at the endpoints of the interval.

9. Find the absolute maximum and absolute minimum values of $f(x) = x^3 - 3x^2 - 9x + 1$ on $[-2, 1]$.

We calculate $f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1)$.

We see that

$$f'(x) = 0 \text{ when } x = -1, 3$$

$$f'(x) \text{ is never undefined.}$$

Therefore, the critical numbers of the function $f(x)$ are $x = -1$ & $x = 3$. Note however, that $x = 3$ is not in the given interval and so we do not consider this value moving forward.

We also consider the endpoints of the given interval $x = -2$ & $x = 1$.

$$f(-2) = -1 \quad \& \quad f(-1) = 6 \quad \& \quad f(1) = -10$$

Therefore, we see that $f(x)$ has an absolute minimum $f(1) = -10$ and an absolute maximum $f(-1) = 6$.

10. One way of proving that $f(x) \leq g(x)$ for all x in a given interval is to show that $0 \leq g(x) - f(x)$ for all x in the interval; and one way of proving the later inequality is to show that the absolute minimum of $g(x) - f(x)$ on the interval is non-negative. Use this idea to prove that $\sin(x) \leq x$ for all x in the interval $[0, 2\pi]$.

We want to see when $x - \sin(x) \geq 0$. Let $f(x) = x - \sin(x)$.

Therefore, $f'(x) = 1 - \cos(x)$.

$$f'(x) = 0 \text{ when } \cos(x) = 1 \rightarrow x = 0 \text{ \& } 2\pi$$

We then test

$$x = 0, \quad f(0) = 0 - \sin(0) = 0$$

$$x = 2\pi, \quad f(2\pi) = 2\pi - \sin(2\pi) = 2\pi$$

Therefore, the absolute minimum of $f(x)$ on $[0, 2\pi]$ and so $x - \sin(x) \geq 0$ on $[0, 2\pi]$ which implies that $x \geq \sin(x)$ on $[0, 2\pi]$.

11. The concentration $C(t)$ of a drug in the bloodstream t hours after it has been injection is commonly

model by the equation $C(t) = \frac{K(e^{-bt} - e^{-at})}{a - b}$, where $K > 0$ & $a > b > 0$. At what time does the maximum concentration occur?

$$\text{We calculate } C'(t) = \frac{K}{a - b}(-be^{-bt} + ae^{-at})$$

We see that $C'(t) = 0$ when

$$\begin{aligned} -be^{-bt} + ae^{-at} &= 0 \rightarrow ae^{-at} = be^{-bt} \\ &\rightarrow \ln(ae^{-at}) = \ln(be^{-bt}) \\ &\rightarrow \ln(a) - at = \ln(b) - bt \\ &\rightarrow at - bt = \ln(a) - \ln(b) \\ &\rightarrow t = \frac{\ln(a) - \ln(b)}{a - b} \end{aligned}$$

$C'(t)$ is never undefined.

Note that as $x \rightarrow \infty$, $C(t) \rightarrow 0$.

Note that $C(t)$ is always non-negative.

Based on this, and the fact that we only have one critical point, our critical value must correspond to an absolute maximum.